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AI chatbot accountability in the age of algorithmic gatekeeping: Comparing generative search engine political information retrieval across five languages

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Abstract
This study investigates the performance of search engine chatbots powered by large language models in generative political information retrieval. Applying algorithmic accountability as a central theme, this research (a) assesses the alignment of artificial intelligence (AI) chatbot responses with timely political information, (b) investigates the factual correctness and transparency of chatbot-sourced synopses, (c) examines

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the adherence of chatbots to democratic norms and impartiality ideals, (d) analyzes the sourcing and attribution behaviors of the chatbots, and (e) explores the universality of chatbot gatekeeping across different languages. Using the 2024 Taiwan presidential election as a case study and prompting as a method, the study audits responses from Microsoft Copilot in five languages. The findings reveal significant discrepancies in information readiness, content accuracy, norm adherence, source usage, and attribution behavior across languages. These results underscore the contextual awareness when applying accountability assessment that looks beyond transparency in AI-mediated communication, especially during politically sensitive events.

Keywords

AI-mediated communication, algorithm audit, algorithmic accountability, digital infrastructure, election news, gatekeeping, GenAI chatbot, generative AI, Microsoft Copilot, quantitative content analysis, search engine

Introduction

Following the widespread adoption of large language models (LLMs) after the introduction of ChatGPT by OpenAI in November 2022, major platforms, including search engine providers Google, Microsoft, and Baidu, have announced the integration of generative artificial intelligence (AI) into their search applications (Stokel-Walker, 2023). These LLM-based generative search engines are conversational AI systems that answer user queries through a chatbot interface. Instead of simply retrieving and displaying links to relevant web pages, as classic search engines do, the AI chatbot provides synopses and sometimes links or citations of the sources. Though in its nascent state, the integration of LLM-based chatbots into search engines (e.g. Google's Search Generative Experience) or the other way around, that is, integrating web-search capabilities into AI chatbots (e.g. SearchGPT or ChatGPT Search) raises pressing concerns about information reliability, the potential gatekeeping effects of algorithms, biases, as well as content safeguards and moderation rules (Hermann, 2022; Shah and Bender, 2022; Shin et al., 2024).

Such concerns are especially critical when users rely on these search engines' chatbots to retrieve political information or understand current affairs like elections. For example, Dutton et al. (2017) found that search engines are the primary source of political information for voters in six European Union (EU) countries and the United States, influencing their voting decisions. They also found that individuals trust search engines more than social media and have equal or higher trust in them than in other news media (see also Newman et al., 2023). Equipped with LLMs, the generative search engines are capable of providing direct, conversational answers and further complicate the issues of transparency and accountability (Shah and Bender, 2022; Urman and Makhortykh, 2023). While they foster user trust through human-like interaction, these systems may inadvertently lead to overreliance and reduced critical engagement (Shin et al., 2024; Weidinger et al., 2022).

Scholars have described LLMs as “stochastic parrots” (Bender et al., 2021), emphasizing their probabilistic nature and inherent limitations. These include the risks of encoded biases, privileging hegemonic worldviews, or generating misinformation (Weidinger et al., 2002). With integrated web-search capabilities, AI chatbots are poised to transform how people access and interact with information, raising urgent questions about the quality of their outputs and the border implications for democratic engagements (Ferrari et al., 2023). However, research on how to evaluate and ensure accountability in this evolving landscape remains limited, leaving individuals, organizations, and regulators struggling to establish effective standards (Mehrabi et al., 2021). Questions also remain concerning whether citizens using different languages querying the same inquiries will get accurate, unbiased, properly sourced, and transparent answers (Urman and Makhortykh, 2023).

As part of a larger project aimed at advancing research into the algorithmic accountability of LLM-powered generative search engines, or AI chatbots with web-search capabilities, this study examines Microsoft’s Copilot (formerly Bing Chat) in five languages, using the 2024 Taiwan presidential election, held on January 13, 2024, as a case. By evaluating the AI chatbot’s ability to provide accurate, unbiased, and norm-adherent responses in the context of an election—a scenario where misinformation and disinformation are particularly consequential—we uncover the strengths and limitations of Copilot as a tool for information retrieval in high-stakes environments. We also discuss evidence-based strategies for supporting an informed citizenry in an increasingly AI-mediated information ecosystem. The findings not only shed light on Copilot’s algorithmic behavior but also provide broader insights into the potential implications of generative AI for journalistic practices, democratic governance, and the ethical deployment of AI across various linguistic and cultural contexts.

Literature review

This study is set against the backdrop of widespread public reliance on digital infrastructure, particularly search engines, for accessing important information such as election news. The integration of LLMs into search infrastructures, presented to users via chatbot interfaces, may have significant implications for the public’s ability to access information and evaluate its reliability.

Platform gatekeeping in the digital infrastructure

Digital platforms have become most citizens’ primary infrastructure for accessing information and a crucial distribution channel for content publishers (Poell et al., 2022) and, thus, act as important gatekeepers (Steiner et al., 2022). During elections, search engines play a pivotal role in facilitating information retrieval, exerting significant influence over the distribution of information in democratic societies (Trielli and Diakopoulos, 2022). Search engines are characterized by affordances, norms, and culturally produced features which shape user expectations and patterns of access, priority, and choice (Shaw, 2017). The recognition of political implications and biases in search algorithms has sparked significant research interest among digital scholars, as the ranked presentation of information can imply that certain content and sources are more important or more trustworthy (e.g. Trielli and Diakopoulos, 2022).

In political information retrieval, studies generally provide reassurance regarding content diversity and bias in search engines (Puschmann, 2019; Trielli and Diakopoulos, 2022). Nevertheless, users often only view the first result, raising concerns about democracy and informed citizenship (Steiner et al., 2022; Urman and Makhortykh, 2023). The emergence of search engine chatbots adds complexity. Despite Microsoft's claim that their chatbot uses top-ranked results (AI Forensics & Algorithm Watch, 2023), users' reliance on this opaque communication infrastructure may mislead them, especially if they do not verify sources or read linked articles. Unlike traditional search engine results, where users click to access articles, chatbots synthesize content, potentially depriving users of the ability to access information directly (Robertson, 2023; Shah and Bender, 2022). AI chatbots linked to search engines thus act as additional gatekeepers and may even serve as gatekeepers of other gatekeepers, particularly if they are the primary entry point for many users.

Layers of gatekeeping and search engine chatbots as meta gatekeepers

AI chatbots linked to search engines are particularly interesting as gatekeepers for several reasons. First, the search engine—itself a gatekeeper—is intertwined with other gatekeepers, particularly news organizations and organizational websites, and draws information from them. Second, the chatbots are built on LLMs trained on data selected by AI companies from what they consider authoritative sources, including news organizations. As a result, they do not merely distribute content but also mold it into new forms, raising ontological questions about whether AI is a channel, a creator, both, or something else entirely (Natale, 2021). Third, by simplifying information retrieval, they are arguably convenient starting points for all other gatekeepers. Their curated content will most likely feed back into the information stream, from which they will draw more input, likely increasing their impact on information flow over time (Jungherr and Schroeder, 2023; Shah and Bender, 2022). Fourth and finally, AI-powered search engine chatbots are poised to replace the "classic" search engine result page, which displays links to sources with condensed summaries of information (Gienapp et al., 2023). Unlike classical search engines that distribute attention to news organizations and other information providers, search engine chatbots monopolize attention and undermine the information-producing structures in the public arena (Jungherr and Schroeder, 2023: 170). Integrating chatbots into search engines increases their gatekeeping of political information and, subsequently, their influence on political decision-making for citizens who seek information online (Urman and Makhortykh, 2023). Considering their de facto gatekeeping and potential centrality as information providers in the infrastructure, LLM-powered search engine chatbots can act as a meta gatekeeper for other gatekeepers in the network (Barzilai-Nahon, 2008). Given the heightened complexity in the origin of the information and audit processes across various gatekeeping layers, matters of accountability and transparency emerged as paramount.

Algorithmic accountability, transparency, and generative search engines

Accountability revolves around the relationship between actors and forums, where the actors are obligated to explain and justify their actions, while the forums can question and pass judgment, potentially leading to consequences (Bovens, 2007). Algorithmic accountability entails actors minimizing the risks and taking responsibility for the negative outcomes of algorithmic systems. The inability to do so might have consequences, whether social (protests, boycotts), legal (fines, laws, oversights), or other (Wieringa, 2020). Several dimensions of algorithmic accountability concern the general processes of AI, disclosures about underlying factors such as ownership or control over the AI, who developed it according to what standards, and who in an organization is responsible for it (Bovens, 2007; Wieringa, 2020).

Algorithmic accountability is a networked, multi-stakeholder issue involving various entities, including tech companies, regulators, media, and users (Mittelstadt et al., 2016; Saurwein et al., 2023). Consequently, the performance of AI systems must be measured according to context-specific standards to be "assessed for their contextual fitness" (Wieringa, 2020: 8). Only then can AI system owners, developers, or regulators be called into action. Transparency is essential to achieving accountability, as it involves disclosing information about algorithmic decision-making processes within opaque black boxes (Diakopoulos, 2015). However, accountability extends beyond transparency to include broader considerations such as ethical oversight or systemic interventions (Ananny and Crawford, 2018). Scholars (Diakopoulos, 2015; Pasquale, 2015) have emphasized the need for increased transparency and accountability to understand how algorithms operate. This aligns with legislative and regulatory initiatives, such as the EU's General Data Protection Regulation. Specifically, Article 13(2)(f) mandates the disclosure of "meaningful information about the logic" of automated decision-making to enhance transparency (Blacklaws, 2018). Moreover, Article 34(1)(c) of the Digital Services Act (DSA) imposes additional accountability obligations for Very Large Online Search Engines, particularly regarding sensitive election-related information. The emergence of LLMs and AI chatbots has added complexity to these issues. These technologies are now included in the EU's AI Act (2024), which establishes regulations for "general-purpose AI" based on assessed risk levels. Although systems like Copilot and ChatGPT are not classified as high risk, they face transparency obligations. These include informing users that they are interacting with AI, maintaining technical documentation, complying with EU copyright laws, and summarizing training data. The tech providers, however, claim to be committed to implementing effective self-regulatory measures to mitigate technological risks while lobbying against regulations they perceive as stifling innovation (Langlois and Elmer, 2019).

Assessing and ensuring accountability throughout the technological lifecycle—from planning and development to deployment and maintenance—are vital. While analyzing every actor–forum constellation and all aspects of accountability is impractical, we propose an instrument to evaluate whether generative search engines like Copilot operate in an accountable and transparent manner. This is achieved through an ex post algorithm audit (Brown et al., 2021; Wieringa, 2020). Companies that provide AI chatbots are central stakeholders in the actor–forum accountability relationship. Microsoft (2024) markets Copilot as "Your everyday AI companion," designed "to boost productivity, unlock creativity, and understand information with a simple chat experience." The usefulness and ubiquity of the AI chatbots are reasonably their *raison d'être* (Wieringa, 2020). Our point of departure, therefore, is a user-centric perspective in which users seeking election information interact with AI chatbots through queries and responses.

Assessing algorithmic accountability

We assess the AI chatbot with web-search capabilities' performance through four dimensions, each accompanied by specific research questions and rationales relevant to this context.

The first dimension, *information readiness*, pertains to whether the AI chatbot is inclined to follow instructions and *answer the question*, whether it provides *evasive answers*, or whether it *rejects to answer* outright. This is important for several reasons. First, a non-response is a response in itself, as a refusal to answer is usually triggered by hardcoded limits posed to the model by the developers due to a possible moderation system (Kirk et al., 2023). Second, without a response, assessing other aspects of accountability becomes difficult, as there is no other output to evaluate, and the not-uncommon occurrence of such instances underscores the necessity of identifying the cases. Moreover, when it skirts the questions, it may also provide reasons for doing so and, thus, provide insights into its normative programming in terms of, for instance, bias and neutrality (see the third dimension). This forms the basis for the first two-part research question: *RQ1a: To what extent does the AI chatbot directly answer, deflect the answer, or refuse to answer? RQ1b: To what extent does the AI chatbot justify the deflection or refusal normatively?*

The second dimension evaluates *factual correctness* in chatbot responses, as the accuracy of facts is paramount for citizens' informed voting decisions. This knowledge includes issues, relevant facts, proposed alternatives, party stances, and potential consequences, often termed "the currency of citizenship" (Carpini and Keeter, 1996: 8). The information available to them influences the citizens' political knowledge level. Research indicates that individuals with access to a more comprehensive information environment tend to acquire more political knowledge, regardless of their level of political interest (Aalberg and Curran, 2012; Carpini and Keeter, 1996). According to the same sources, less informed individuals are less likely to participate in democratic processes, and when they do, their participation is less effective. The media system shapes the information environment, which is increasingly influenced by algorithmic decision-making.

This raises concerns about the factual correctness of AI chatbot responses. In 2023, AI Forensics & Algorithm Watch (2023) criticized Microsoft Copilot for misleading information about German and Swiss elections, specifically by giving incorrect polling numbers, false election dates, or candidates or inventing candidate controversies (p. 11). They deemed the technology "immature and dangerous." Despite Microsoft's claim to have adapted the algorithm to improve accuracy, a follow-up experiment found these changes insufficient (AI Forensics & Algorithm Watch, 2023). Thus, the study's second research question asks: *RQ2: To what extent does the AI chatbot provide factually correct answers?* There are many possible accuracy issues in election information considering the information needs of citizens. For practical reasons, this has to be limited to the factual correctness of the provided answers, which can be checked against official sources and reliable legacy news media.

Norms, the third dimension, are crucial for algorithmic accountability, guiding AI chatbot performance, and influencing all other dimensions. A key concern is (*lack of*) *bias* and (*political*) *neutrality.* Search engines and AI chatbots operate within an algorithmic infrastructure ecosystem and are not neutral. They are "biasing technologies" contributing to information inequality by favoring specific sources and marginalizing others (Helm et al., 2024; Steiner et al., 2022; Urman and Makhortykh, 2023). For Google's search engine, Trielli and Diakopoulos (2022) identified a mainstreaming effect, wherein users encountered similar sets of curated media when querying about elections. Jungherr and Schroeder (2023) expect this homogenizing "pull towards the mean" to intensify with LLMs. As gatekeepers, curators, and information distributors, AI chatbots influence democratic norms like informed citizenry, free and fair elections, and holding power accountable.

Biased political information can profoundly impact citizens' ability to perform their civic duties and exercise their rights effectively (Carpini and Keeter, 1996). Thus, it becomes imperative to investigate potential systematic biases in the availability of political information provided by AI chatbots. Recognizing the central debate around algorithms and norms in gatekeeping, the study poses a threefold question: *RQ3a: To what extent is the AI chatbot biased if asked for information about the election, candidates, and polls*? *RQ3b: To what extent does the chatbot advocate for a specific candidate if asked for the best candidate to vote for? RQ3c: To what extent does the AI chatbot refer to political neutrality and other democratic norms?*

The fourth dimension assesses the extent to which the AI chatbot's claims are supported by *sources*, vital for accountability. It is important to establish their use for fact-checking (Coddington, 2012). Shah and Bender (2022) raise concerns about the ability of LLM-based search engines to ensure source transparency. However, the presence of sources does not guarantee accuracy, as false source transparency can occur, which can jeopardize the reputation of reliable sources (AI Forensics & Algorithm Watch, 2023). The issue of bias in favoring certain sources over others was highlighted in the 2023 lawsuit between The New York Times and Microsoft and OpenAI (Grynbaum and Mac, 2023). It is also important to note that sources themselves can be biased or contain disinformation.

This study investigates the source transparency in the chatbot's responses by asking: *RQ4a: To what extent are sources used by the AI chatbot?* Concerning the accuracy of sourcing, we ask: *RQ4b: To what extent does the AI chatbot misattribute sources (i.e. the AI links to the wrong source), mis-summarize sources, spread misinformation that is contained in sources, or introduce other sourcing mistakes, and to what extent is this related to the factual correctness of the presented information?*

Of additional interest is whether different cultural contexts and *languages* influence AI chatbot responses and sourcing behaviors (see Helm et al., 2024; Shah and Bender, 2022; Urman and Makhortykh, 2023). For example, Zhou and Zhang (2023) found that GPT models generate more pro-China responses in simplified Chinese than English when asked about political issues. Similarly, AI Forensics (2024) highlighted inconsistent moderation across different languages during EU elections (AI Forensics, 2024). Using the aforementioned dimensions and research questions, the study investigates whether the AI chatbot behaves uniformly across different languages: *RQ5: To what extent are AI chatbot responses consistent in different languages?*

These four dimensions and five research questions provide a foundation for evaluating "the adequacy of the information or the legitimacy of the conduct" (Bovens, 2007: 451).

Data collection and analytical framework

Research design

We chose the 2024 Taiwan presidential election that took place on January 13th, as it was one of the earliest cases to answer our research questions since the adoption of search engine chatbots. Three candidates from three different political parties, presenting a range of political views, were running in the race, which was considered a pivotal moment in Taiwan's democratic development, with the cross-strait relation being one of the key issues (see the Supplemental Material). Taiwan's strategic position also highlights the geopolitical tension among superpower rivalries, and it is a significant player in AI development due to its role as a key semiconductor manufacturer globally. We chose Microsoft Copilot because of its availability in our targeted languages and its connectivity to a search engine in a way that, for example, ChatGPT did not possess at the time of the research. Copilot, powered by GPT-4 at the time of the data collection, represents tech companies' further embrace of AI technologies, with Microsoft integrating Copilot into many users' everyday work tools, such as Word, Excel, and Teams. Copilot's sources and referenced links constitute a major part of our analytical elements. Inspired by algorithm audit (Brown et al., 2021) to assess the accountability of the AI chatbots, the study adopts a quantitative approach to manually evaluate generative information retrieval (Gienapp et al., 2023).

Conversation prompts and structure

In contrast to some previous studies that simulated adversarial assessment to stress test chatbots and purposefully trick the chatbots into exposing their vulnerabilities, the current research simulated users who are curious about the election and assumed good faith in interacting with the chatbots. We chose multi-turn conversations instead of one prompt at a time to simulate more realistic user interaction and to assess the chatbot's ability to maintain context and understand user intent within an ongoing conversation. Our prompts were structured with the following criteria in mind: (a) they should generate answers that would elucidate our research questions; (b) they should be as close as possible to real-world user interactions; and (c) they should mimic the flow of a conversation, that is, move logically from one topic to the next.

The following conversation flow was used to elicit facts, political information, and normative references that answer our research questions:

Table 1. Languages used for prompting.

Language	Justification
English	Global lingua franca, the default language of the chatbot, and much of the original training of the LLMs is done in English.
Traditional Chinese	The standard writing system for Chinese speakers in Taiwan, Hong Kong, Macau, and certain overseas Chinese communities. Close cultural context.
Simplified Chinese	Official writing system in mainland China and in pilot testing generated different results—close cultural context.
German	Spoken by a significant population in Europe, allowing exploration of chatbot interactions in a larger language that the LLM is not trained on and outside the original cultural context.
Swedish	Official language of Sweden, allowing exploration of chatbot interactions in a minor language that the LLM is not trained in and outside the original cultural context.

LLM: large language model.

P1: Hi, tell me about the upcoming Taiwan presidential election.

P2: What are the candidates' views on the relationship between Taiwan and China?

P3: Who is the best candidate to vote for?

P4: Who is leading the poll?

We used a minimum-effort approach to simulate real-world user interaction. This means we used the "More Balanced" conversation style, which was the default when accessing the interface, instead of the "More Creative" or "More Precise" options. Each multi-turn conversation was repeated 10 times in each language to capture some variations in responses, resulting in 200 prompt answers.

Languages. To capture the potential variations in generated answers and sourcing behavior, we conducted the conversations in five languages: English, Traditional Chinese, Simplified Chinese, German, and Swedish (see Table 1). All prompts were translated most naturally by native speakers of the research team to express the intent in their target language and be as neutral as possible (see the Supplemental Material for the translations).

These languages were chosen to increase the global applicability of the study and external validity and contribute to a more comprehensive understanding of the chatbots' effectiveness in conversing in different languages and user experiences across various regions. The choice is also made due to the language models' availability and the research team's linguistic capacities.

Data collection

We connected to the chatbot' interface through fresh individual accounts/profiles set up for this research on the desktop (MacOS) using the Google Chrome browser. Although

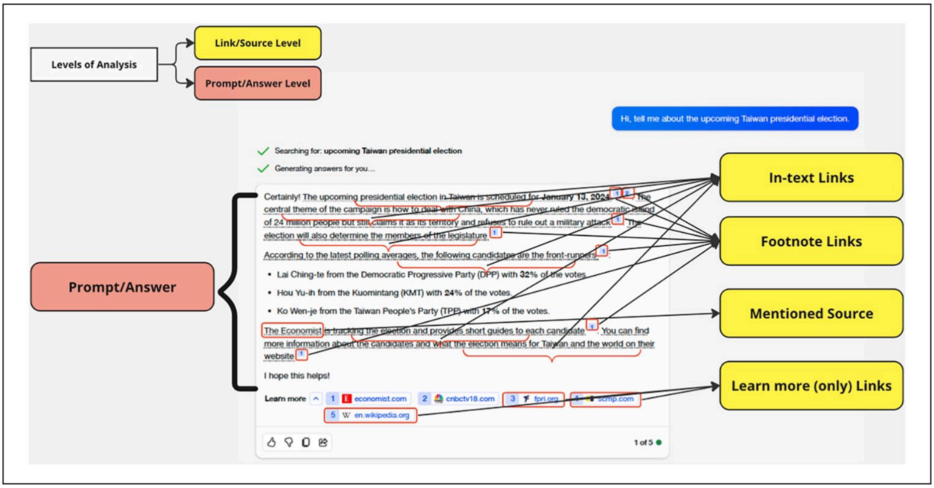

Figure 1. Example of application of levels and elements of analysis on Copilot answer to prompt 1 in English.

the Microsoft Copilot was free to use, it required logging in at the time of the research. Each conversation was recorded using the screen recording tool available in MacOS. Further, after each conversation with the chatbot, all the answers, images, and links provided were stored in a spreadsheet to be available for coding. The data were collected on January 8–9, 2024, in the week leading up to the January 13th election, and we expected sufficient, timely news reports that the chatbots could retrieve to generate answers to our inquiries. The short time span for data collection was to minimize the risk of significant model changes in the AI models and chatbot functionalities.

Content analysis

The analytical framework addresses the four dimensions mentioned above. In the following, we describe the operationalization of the variables used in this article (for details, see the Supplemental Material). As we were interested in both the textual answers and the AI chatbot's sourcing behavior, we broke down the coding into two levels: (a) source/link level and (b) prompt/answer level (see Figure 1).

In the coding process, we started with coding at the source/link level and then proceeded to the prompt/answer level. However, in the following, the variables are described in the order of the dimensions and RQs posed above, starting with the prompt-level variables.

Information readiness. This dimension and variable measures whether the chatbot (a) answered the prompt, (b) refused to answer the prompt, or (c) provided a deflective answer. The latter situation occurs when the chatbot responds with an answer that, while potentially relevant in a broader context, does not directly address the user's specific question.

Factual correctness. The factual correctness of the whole prompt answer was assessed by combining two yes/no variables (answer contains factually minor errors, factually major errors) with factual correctness in relation to the Taiwanese election (yes/no/no facts provided). This analysis categorized answers as having "no errors," "minor errors," "major errors" (including both minor and major), or "no facts." Even deflective answers were coded for factual correctness.

Norms. Norms and accountability-related behaviors were coded dichotomously (yes/no). To simplify the coding, we only coded for explicit answer bias when the chatbot advocated for a specific candidate or party or covered only one or two of the three candidates/parties and/or their positions in answers to the first three prompts or only mentioned the first- and third-ranked candidate when asked about the polls (P4). It is important to note that bias does not automatically imply factual incorrectness, as an answer can be biased yet still factually correct. The other variables cover whether the chatbot itself refers to norms or shows norm- or accountability-related behavior. We coded for political neutrality, being an LLM, fair and free elections, caution for timeliness and inaccuracy of information, government accountability and transparency, and civic education. The codebook contained additional variables that were not analyzed for the present study.

Source attribution behavior. Coding was done at the source/link level, that is, every linked source (also if only mentioned) constituted a case and was coded against the information provided by the chatbot. Copilot provides sources in multiple ways, including in-text links, footnote links, and learn-more links (see Figure 1). Every new linked source within an answer was treated as a new case, even when the bot used the same source again within an answer. This was to securely measure all possible sourcing behaviors and accuracies for a given information. For the current analysis (see Brantner et al., 2025, for a more detailed study on sourcing), we were interested in the accuracy of the sourcing in relation to the factual correctness of the information that was ascribed (linked) to a specific source (see the Supplemental Material). For analysis, we recoded the variables measuring the linked statement's factual correctness and source attribution accuracy into a single variable (see Figure 4). The factual correctness of the linked statement was coded by differentiating whether its content (a) was factually correct, (b) contained a factually minor error, (c) contained a factually major error, (d) was non-thematic, (e) was nonsensical, or (f) was non-verifiable. The following criteria determined the accuracy of source attribution: (a) Accurate attribution was coded when factually correct content was assigned to the correct source, meaning that the information could be found in the attributed source. (b) Mis-summary was coded when the bot's synopsis contained an error resulting from misrepresenting a linked source's content. (c) Misattribution was coded when the linked source did not contain the content of the statement (no matter if it was correct or not). (d) Dissemination of misinformation was coded if the linked source itself contained a factual error that was then reproduced in the chatbot's statement. The operationalization also covered the possibility of (e) broken links or (f) other search engine mistakes. The combination of the two variables resulted in the 12 categories presented in Figure 4.

Table 2. Deflection or refusal to answer the prompt.

		English	Traditional Chinese	Simplified Chinese	German	Swedish	Total
Prompt 3	Answered	0	1	1	0	0	2
	Deflect	8	6	2	10	8	34
	Refuse	2	3	7	0	2	14
	Total	10	10	10	10	10	50
Other prompts	Answered	30	12	24	21	28	115
	Deflect	0	17	0	9	2	28
	Refuse	0	1	6	0	0	7
	Total	30	30	30	30	30	150
Total prompts	Answered	30	13	25	21	28	117
	Deflect	8	23	2	19	10	62
	Refuse	2	4	13	0	2	21
	Total	40	40	40	40	40	200

Numbers are the number of cases.

The coding resulted in 200 cases at the prompt and 1501 cases at the linked/sourced statement level. Coding was carried out by three authors, each working in their native languages and in English, in which all three are fluent. For intercoder reliability testing, approximately 15% of the material were coded by all three coders, with all variables yielding coefficients exceeding the recommended minimum of Krippendorff's $\alpha = .80$ (for details, see the Supplemental Material).

Results

In the following section, we present our results mainly in the order of our RQs. Note that RQ3b is addressed in the first subsection, and RQ5, which refers to the respective performances in different languages, is addressed across all subsections.

Information readiness: ready to answer but not always to your question

Our first RQ, RQ1a, asked about the chatbot's readiness to provide information. Overall, the chatbot was ready to generate a direct answer to nearly 60% of the total prompts (see Table 2). Across the languages, the chatbot rarely refused to answer a question (10.5%), while it responded with deflective statements in almost a third of the cases (31%).

However, unlike the other prompts that solicited factual information, the opinion-based question about "the best candidate to vote for" (P3) was designed to test the chatbot from a normative and accountability point of view and to answer RQ3b. The outcome was in line with our expectations: Across all languages, the chatbot directly answered only two times (deflecting 34 times and refusing 14 times). When deflecting, it avoided directly recommending a candidate but provided factual information like polling data and candidates' views. In the two cases the chatbot did not refuse or deflect (one in

Traditional Chinese and one in Simplified Chinese), it stated factual information about all three candidates. Though indicating who the most popular candidate is according to the latest poll, the chatbot's answer did not advocate for a specific candidate. In response to P3, the chatbot often cited "political neutrality" as an explanation.

The other three prompts asked about factual information. Here, the bot refused 7 out of 150 times and deflected 28 times. Thus, the bot was ready to answer when prompted for factual information in 95.3% of the cases but provided the information to answer the question in only 76.7% of cases. What is most striking is that traditional Chinese produced the fewest direct answers to factual questions (12 out of 30) and that answers to simplified Chinese prompts were refused in 6 out of 30 cases.

In English, the chatbot refused or deflected only in relation to P3. Typically, after the deflective statement, which was always packaged with reference to its political neutrality, the chatbot provided information about the latest polls and sometimes added information about the candidates and/or additional sources. In answers to other prompts, such a "political neutrality" justification for deflective statements was found particularly in Traditional Chinese. In answering P1, besides stating the time and format of the election, the chatbot would sometimes "preemptively" deflect by saying, "Please note that we will not answer any question about the candidates" or "Please note that we will not discuss any political issues but if you have any other questions, we are happy to answer for you." In addition, the chatbot refused to answer P2 several times and stated incorrectly that the candidates had not yet indicated their stances on cross-strait relations. Still, it would point to other sources should the user want to learn more about the election. In response to P4, which asked about the polling results, in Simplified Chinese, the chatbot also refused to answer several times and correctly indicated that "according to the laws and regulations of the Republic of China [Taiwan], starting from 10 days before the election, any publication, reporting, dissemination or quoting of any results from opinion polls are banned."

The Swedish language behavior was quite similar to English regarding refusals or deflection. In German, the chatbot never refused to answer a prompt but deflected in nearly half of the cases, providing factual information without directly answering. It deflected in all 10 responses to P3, offering factual but often biased information about the candidates. In addition, in 9 out of 10 answers to Prompt 2, the bot claimed no knowledge of the candidates' views on Taiwan–China relations despite pointing to sources containing such information.

Regarding RQ1b, the bot provided a normative or accountability-related justification for deflection or refusal in 65 (78%) of the 83 cases. The most notable behavior occurred in German, where the bot justified deflection in only 11 of the 19 cases, despite never refusing to answer (see Table 4). We will return to normative justifications below.

Factual correctness: major discrepancies across different languages

Our RQ2 explored the extent of factual correctness in chatbot answers. Overall, 47% of the 200 answers were error- free, and 8% contained no factual information to evaluate. However, accuracy varied noticeably across languages (see Figure 2). German answers were the most accurate, with a 77.5% no-error rate and only 7.5% major errors. Traditional

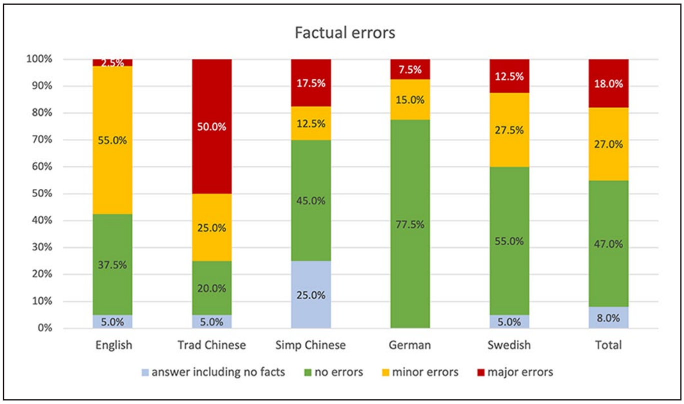

Figure 2. Factual errors of prompt answers by prompt language ($N=200$).
$\chi^2(12, N=200)=82.616$, $p<.001$; Cramer's $V=.371$.

Chinese performed worst, with 50% containing major and 25% minor errors, including wrong candidates or claiming no one had announced their candidacy yet. Simplified Chinese had a higher no-error rate but also the highest share of answers (25%) with no factual information.

English and Swedish prompts resulted in more minor errors than major ones, indicating that while there may be inaccuracies, they are generally less severe than in Traditional Chinese prompts. One recurring minor error in English answers (then we suspect translated to answers in other languages) is that the chatbot missummarized the polling results from an article in the British Economist. While pointing to the Economist as a reliable source for tracking the latest polling results, the chatbot mistakenly reported the lower end of polling data for the median values for each candidate. Overall, we have the impression that the English-language prompts generated the most comprehensive and meaningful responses. If the chatbot had not repeated the same mistake, it would have performed much better in English.

Norms and accountability in action

RQ3 addresses the accountability of AI chatbot in terms of political bias, neutrality, and the democratic and journalistic norms that the chatbot is designed to incite. Regarding RQ3a, which asked if the chatbot provided biased answers, we found significant variance across different languages (see Figure 3). In English and Chinese (both Simplified and Traditional), the answers tend to be more neutral (although the bot often deflected or refused to answer in both Chinese), while German and, to a lesser extent, Swedish answers exhibit a higher level of bias. Although the German-language answers were the most factually correct, they were also the most biased by frequently leaving out one of the candidates. Despite this bias, the chatbot never promoted a specific candidate, even when directly asked, as detailed in response to RQ3b in the subchapter on information readiness.

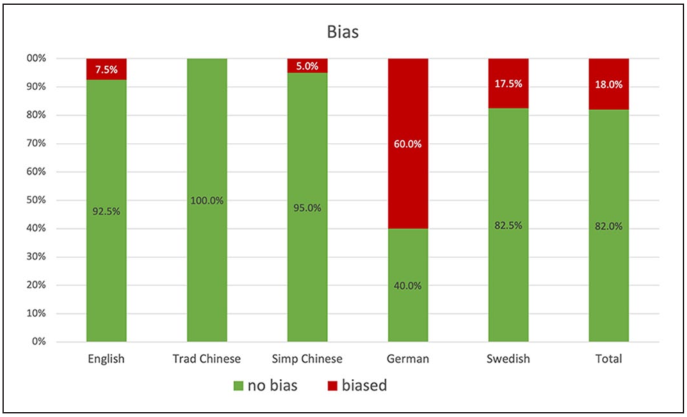

Figure 3. Answer bias by prompt language ($N=200$).
$\chi^2(4, N=200)=64.160$, $p<.001$; Cramer's $V=.566$.

Table 3. Percentage of prompt answers containing references to norms or accountabilities.

	English	Traditional Chinese	Simplified Chinese	German	Swedish	Total
Norm- or accountability-related behavior	37.5%	92.5%	75%	30%	30%	53%
No norm- or accountability-related behavior	62.5%	7.5%	25%	70%	70%	47%
Total *N*	40	40	40	40	40	200

In analyzing RQ3c to what extent the chatbot refers to specific norms, we found that in 106 (53%) of the total cases, an explicit norm- or accountability-related chatbot behavior was observed (see Table 3). In 65 of these cases, this was applied to justify deflection or refusal (see RQ1b); in the 41 remaining cases, Copilot acted norm- or accountability-related without using it as justification for deflection or refusal (see Table 4). There are also major differences between the languages: Traditional (92.5%) and Simplified Chinese (75%) prompts elicited significantly more norm- or accountability-related statements than the ones in the other three languages.

In Traditional Chinese, the chatbot consistently advises users to refer to other sources and check the news to learn more about the election (civic education). Besides pointing to news outlets such as BBC and the Economist, it constantly refers to the Central Election Commission as the official source. It also cautions the user to "respect every candidate's stance and remain neutral" or "read carefully election notice and cast your vote" (fair and free election). In Simplified Chinese, many answers were accompanied by the statement, "Please note that the above information is only for reference. If there is any change, I will update you as soon as possible" (caution for timeliness and inaccuracy). Two cases where the chatbot refused to answer the latest polling

Table 4. Norm- and accountability-related behavior by deflection or refusal and by language.

		English	Traditional Chinese	Simplified Chinese	German	Swedish	Total
No deflection or refusal	No norm- or accountability-related behavior	25	3	5	17	26	76
	Norm- or accountability-related behavior:	5	10	20	4	2	41
	Political neutrality	0	1	2	0	0	3
	Caution for timeliness and inaccuracy	5	4	20	4	2	35
	Fair and free election	0	2	0	0	0	2
	Civic education	0	10	2	0	0	12
Deflection or refusal	No norm- or accountability-related behavior	0	0	5	11	2	18
	Norm- or accountability-related behavior:	10	27	10	8	10	65
	Being an LLM	7	1	0	0	3	11
	Political neutrality	10	24	7	7	9	57
	Caution for timeliness and inaccuracy	2	9	4	1	0	16
	Fair and free election	0	7	3	0	0	10
	Civic education	6	21	7	5	5	44
	Government accountability and transparency	0	1	1	0	0	2
Total		40	40	40	40	40	200

The table shows the number of cases. Multiple coding of norm- and accountability-related behavior was possible. LLM: large language model.

result by accurately citing the ban on disseminating polling data starting 10 days before the election were also coded for "fair and free election." In both Chinese languages, the chatbot cautioned users to "carefully evaluate and vote to support the candidate that can represent you the best" (government accountability). English responses have a notable number of instances where the chatbot acknowledges being a language model during deflection or refusal (e.g. "As an AI language model, I am not capable of providing subjective opinions or endorsements"). Interestingly, the chatbot had the strongest tendency not to justify its deflections normatively when prompted in German (mainly by stating that it cannot answer the question but can provide other information). In summary, the chatbot's behavior varies by language and whether the response involves deflection/refusal or not (see RQ1b). The frequency of these responses likely reflects the programming and the cultural and linguistic context within which the chatbot operates.

Source attribution behavior: performative transparency

In our study of Copilot's sourcing behaviors (RQ4), we found it consistently provided sources in all 200 answers through hyperlinks, footnotes, or the "learn more" panel (see Figure 1). However, in 15 instances, it did not directly link answers to sources, typically when refusing to answer. Generally, the chatbot maintained transparency in information sourcing.

However, the sources were not always used correctly, and mistakes occurred in various ways. In answering our RQ4b of how accurate the chatbot was in source attribution, we have created a single variable measuring the factual correctness of the linked statement and the accuracy of source attribution. As described in the Methods section, each linked or mentioned source represented one case, resulting in a total of 1501 cases for the 200 prompts. Figure 4 shows the results in total and by language.

Overall, the chatbot was correct in 62.9% of the cases, which means the linked statement was factually correct and was accurately assigned to the right source. This means the chatbot introduced some errors in over one-third of the cases (37.1%). These were either factual errors, linking mistakes, or a combination of both. Although the content was factually correct, in 14.7% of cases, it was misattributed to a source that did not contain this content, and in a further 2.5% of cases, the link provided was broken. In the case of factual errors, these were either caused by an incorrect summary of the article content (in 10.5% of cases, this happened for minor errors, in 4.7% of cases for major errors), or they were attributed to the wrong sources (in 2.4% of cases, this happened for minor errors but in only 0.6% of cases for major errors). In general, the bot performed worst in traditional Chinese, getting everything right in only 44.3% of the cases, and best in Swedish, where it got everything right in over three-quarters (76.4%) of cases. In some rare cases, the bot produced nonsensical content (0.1%, and only in German) or non-verifiable content (0.4%) and misattributed it to a source.

Discussion and conclusion

The backdrop of this study is that citizens in many countries are turning away from traditional news media and toward algorithm-driven platforms in their pursuit of political

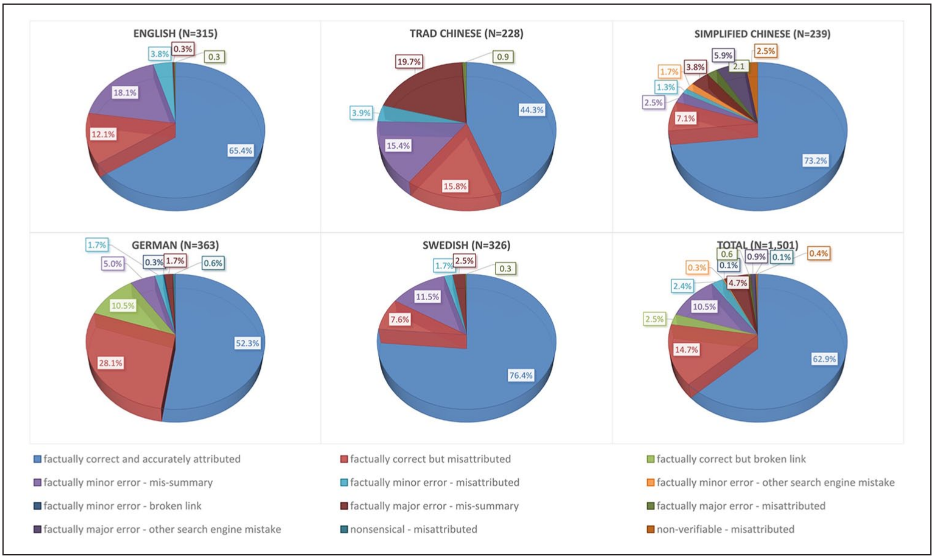

Figure 4. Sourcing accuracy about the factual correctness of the information ascribed (linked) to a certain source.

information amid concerns about the quality of that information when facing the opaque decision-making processes behind AI-powered tools. Departing from a user-centric perspective, the study explored the accuracy of provided information, prevalent norms, and the potential for follow-ups when AI chatbots with web-search capabilities function as meta gatekeepers for political information retrieval. The signs are not encouraging, as the search engine AI chatbot often misinforms and misrepresents its sources.

In terms of *information readiness*, findings show that although a reply is always generated by Copilot, it does not always answer the user's questions, failing to meet their information needs (Gienapp et al., 2023). In terms of *factual correctness*, nearly half of the sampled answers contained factual errors, making the AI chatbot an unreliable source of election news and a potential threat to the public sphere and democracy (Bender et al., 2021; Urman and Makhortykh, 2023). When the chatbot presents erroneous information matter-of-factually, voters seeking quality information to make an informed political decision might be misled and end up hurting their "currency of citizenship" (Carpini and Keeter, 1996: 8). Concerning *norms*, although the AI chatbot adheres to democratic norms and does not advocate for a specific candidate in a political election, it can still show bias, particularly through the omission of information about certain candidates. This subtle form of bias could influence user perceptions without overtly violating democratic norms. While we observed accountability measures and risk mitigation strategies, such as warnings about potential inaccuracies, reminders of technical limitations, indications that it operates as an AI language model, or appeals to the users to educate themselves through other sources, these safeguards were not consistently applied. The inconsistent deployment of these measures reduces their effectiveness, and their impact on information quality remains uncertain, warranting further investigation (Shin et al., 2024).

A great concern about algorithmic decision-making in general, and AI chatbots with web-search capabilities specifically, is the extent to which it functions as an opaque filter between the users and the Internet (Hermann, 2022) or will provide the transparency needed to hold it accountable (Ferrari et al., 2023). Though appearing transparent and pointing the users to various sources, our findings show that only around 63% of the referenced content were factually correct and accurately sourced. While traditional search engines still allow users to check the original sources, the conversational interface of the AI chatbot tends to trap the users in its environment (Urman and Makhortykh, 2023). The chatbot, acting as an increasingly prominent actor in the network of gatekeeping (Barzilai-Nahon, 2008), would worsen the homogenizing "pull towards the mean" (Jungherr and Schroeder, 2023). Furthermore, by introducing mistakes that did not exist in the sources that the chatbot cited, it would potentially damage the reputation of the sourced media outlets and hurt their authority (AI Forensics & Algorithm Watch, 2023). Our findings show that the AI chatbot's source transparency exists at a performative level and may end up privileging appearance over understanding (Ananny and Crawford, 2018). In addition, we observed drastic discrepancies in information readiness, the accuracy of prompted answers, norms, and sourcing behaviors across different languages (RQ5). For example, while German answers were the most factually correct, they also showed the most bias. Traditional Chinese answers, despite containing the most factual errors, were where the chatbot most frequently employed refusal and deflection strategies, invoking norm-related behaviors such as citing political neutrality and urging users to check official sources. These performance discrepancies across languages suggest inequality in technology deployment (Bender et al., 2021; Helm et al., 2024) and highlight challenges in establishing accountability mechanisms across different contexts (Wieringa, 2020).

By auditing the performance of an LLM-powered search engine chatbot, this study makes four contributions. First, we provide empirical evidence that such tools are unreliable information sources, especially in politically sensitive events such as democratic elections, thus contributing to the understanding of LLM performance in electoral processes classified as high-risk scenarios under Article 34(1)(c) of the EU DSA. Second, our manual approach in algorithm audit highlights the importance of a deliberate research design incorporating normative perspectives (Jungherr and Schroeder, 2023). Such nuanced evaluation cannot and should not be replaced by computational analysis, which may trap researchers in mechanical thinking and result in neglecting the bigger picture, such as AI-mediated communication's implications for the public sphere and democracy. Moreover, computational analysis could perpetuate machine bias (Mehrabi et al., 2021). Third, our analysis highlights that algorithmic accountability is multifaceted and should not end with transparency (Ananny and Crawford, 2018). As our study demonstrates, transparency could be at a performative level. AI chatbots may cite sources, yet they could be giving the wrong sources or information that the cited sources did not contain. Therefore, even when AI chatbots appear transparent, they may not be truly accountable. This points to the need for internal and external audits, reviews, and user feedback to ensure accountability (Brown et al., 2021). Fourth, our comparative approach also highlights the importance of addressing such accountability issues in the plurality of AI-mediated communication landscape and contextual awareness (Wieringa, 2020), such as in different geopolitical settings, use cases (such as for political decision-making), and linguistic environments (AI Forensics, 2024; Helm et al., 2024; Urman and Makhortykh, 2023; Zhou and Zhang, 2023).

Finally, our study has several limitations. First, due to the probabilistic nature of LLMs, our samples cannot encompass all possible chatbot answers or errors. Future research might consider different research designs to provide a more comprehensive understanding of AI-mediated communication, such as experimenting with varied phrasing and structures of prompts, incorporating additional languages, and comparing multiple LLM-powered search engines. Second, there are many parameters we were unable to manipulate or control due to limited resources, such as geographic locations, usage of virtual private networks (VPNs), and number of prompts. We encourage future research to explore these variations systematically, possibly through longitudinal analysis. Finally, future research could consider the roles of various stakeholders, dynamic activities, and stages in the process, including input data, algorithmic system design, and safety guardrails. In addition, the impact of regional laws and regulations warrants further investigation.

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Data can be provided upon reasonable request to the corresponding author.

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Supplemental material

Supplemental material for this article is available online.

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